

Wess & Bagger:

• Every rep of susy contains an equal # of b & f

Define: Fermion number op. N_F

$$N_F |b\rangle = |b\rangle, \quad N_F |f\rangle = -|f\rangle$$

$$\{(-1)^{N_F}, Q_\alpha\} = 0$$

Since all eigen states yield $\# \times Q_\alpha$

$$\langle b | \{(-1)^{N_F}, Q_\alpha\} | b \rangle$$

$$= \langle b | (-1)^{N_F} Q_\alpha | b \rangle + \langle b | Q_\alpha (-1)^{N_F} | b \rangle$$

$$= -\langle b | Q_\alpha | b \rangle = \langle b | Q_\alpha | b \rangle$$

$$= -2 \langle b | Q_\alpha | b \rangle$$

Since $\langle b | Q_\alpha | b \rangle$ is positive definite

this implies LHS must equal zero:

$$\{(-1)^{N_F}, Q_\alpha\} = 0 \Rightarrow (-1)^{N_F} Q_\alpha = -Q_\alpha (-1)^{N_F}$$

Now, since $\{Q_\alpha^A, \bar{Q}_{\dot{\beta} B}\} = 2 \sigma_{\alpha\dot{\beta}}^m P_m \delta_B^A$

we would expect $(-1)^{N_F} \{Q_\alpha^A, \bar{Q}_{\dot{\beta} B}\}$ to yield ± 1

$$\text{Tr} [(-1)^{N_F} \{Q_\alpha^A, \bar{Q}_{\dot{\beta} B}\}] = \text{Tr} [(-1)^{N_F} (Q_\alpha^A \bar{Q}_{\dot{\beta} B} + \bar{Q}_{\dot{\beta} B} Q_\alpha^A)]$$

$$= \text{Tr} [-Q_\alpha^A (-1)^{N_F} \bar{Q}_{\dot{\beta} B} + Q_\alpha^A (-1)^{N_F} \bar{Q}_{\dot{\beta} B}]$$

$$= 0$$

$$\therefore \text{Tr} [(-1)^{N_F} 2 \sigma_{\alpha\dot{\beta}}^m \hat{P}_m \delta_B^A]$$

$$= 2 \sigma_{\alpha\dot{\beta}}^m \delta_B^A \text{Tr} [(-1)^{N_F} \hat{P}_m] = 0$$

For fixed momenta, i.e. $\hat{P}_m |n\rangle = P_m^{|n|} |n\rangle$

$$\text{Tr} [(-1)^{N_F}] = 0$$

$$\text{Tr} [(-1)^{N_F}] = \sum_b \langle b | (-1)^{N_F} | b \rangle + \sum_f \langle f | (-1)^{N_F} | f \rangle$$

$$= (\#b) - (\#f) = 0$$

$$\therefore \boxed{\# \text{Bosonic states} = \# \text{Fermionic states}}$$

Reps of SUSY:

(i) Massive reps: $P^2 = -m^2$

Boost to the frame $P_m = (-m, 0, 0, 0)$

$$\{Q_\alpha^A, \bar{Q}_{\dot{\beta}B}\} = 2\delta_{\alpha\dot{\beta}} M \delta_B^A$$

$$\{Q_\alpha^A, Q_\beta^B\} = 0 = \{\bar{Q}_{\dot{\alpha}A}, \bar{Q}_{\dot{\beta}B}\} = 0 \quad \left\{ \begin{array}{l} \sigma^0 = -\mathbb{1}_2 \end{array} \right.$$

Define the ops:

$$a_\alpha^A := \frac{1}{\sqrt{2m}} Q_\alpha^A$$

$$(a_\alpha^A)^\dagger := \frac{1}{\sqrt{2m}} \bar{Q}_{\dot{\alpha}A}$$

$$\{a_\alpha^A, (a_\beta^B)^\dagger\} = \delta_\alpha^\beta \delta_A^B$$

$$\{a, a\} = \{a^\dagger, a^\dagger\} = 0$$

$\Rightarrow \exists$ a "clifford vacuum" $|\Omega\rangle \in \mathcal{H}_{\text{Hilb}}$

$$\text{s.t. } a_\alpha^A |\Omega\rangle = 0$$

• States are built from application of $a^\dagger$ to Ω .

$$\hookrightarrow a_\alpha^A |\Omega\rangle = \Omega_\alpha^A$$

$$\langle \Omega_\alpha^A | \Omega_\alpha^A \rangle \stackrel{!}{=} 1 \leftarrow \text{Normalize states}$$

$$\langle \Omega | a_\alpha^A (a_\alpha^A)^\dagger | \Omega \rangle = -\langle \Omega | \{a_\alpha^A, (a_\alpha^A)^\dagger\} | \Omega \rangle = \delta_\alpha^\alpha \delta_A^A$$

$$\text{We sum over } \delta_\alpha^\alpha = 2$$

$$\Rightarrow \frac{1}{2} \langle \Omega | a_\alpha^A (a_\alpha^A)^\dagger | \Omega \rangle = 1$$

thus, continuing w/ this we find

$$\Omega_{A_1 \dots A_n}^{(\alpha_1 \dots \alpha_n)} := \frac{1}{\sqrt{n!}} (a_{\alpha_1}^{A_1})^\dagger \dots (a_{\alpha_n}^{A_n})^\dagger |\Omega\rangle$$

• Can also think of this as an n -form, w/ normalization half of $\Omega^{(n)}$ since $\langle \Omega | \Omega \rangle = 1$

• Each pair (α_i, A_i) takes $2N$ different values

so n must be: $n \leq 2N$

for any $n \geq 1$, introduce states
the dim of the rep is then given by

Summing over n :

$$d = \sum_{n=0}^{2N} \binom{2N}{n} = 2^{2N}$$

• If Ω is non-degenerate $\Omega^{(N)}$ is the fundamental

irrep. massive multiplet.

$$2^{2N} = 2^{2N-1} \oplus 2^{2N-2}$$

b

f

* Highest spin state: (also the highest weight)

Symmetrize in as many spinor indices as possible

$$\Omega_{A_1 \dots A_N}^{\alpha_1 \dots \alpha_N} : \text{Spin } \frac{1}{2}N$$

* $N=1$: fundamental rep:

Here Ω is the highest weight state

w/ $SU(2)$ subgroup action

$$\begin{cases} \alpha_a^\dagger |\Omega\rangle \\ \frac{1}{\sqrt{2}} \alpha_a^\dagger \alpha_b^\dagger |\Omega\rangle \end{cases} \leftarrow 2 \text{ states } a=1,2$$

$$\frac{1}{\sqrt{2}} \alpha_a^\dagger \alpha_b^\dagger |\Omega\rangle = -\frac{1}{2\sqrt{2}} \epsilon^{ab} (a^\dagger)^\dagger (a_b)^\dagger |\Omega\rangle$$

• $|\Omega\rangle$ has spin 0.

$\alpha_a^\dagger |\Omega\rangle$ has spin $1/2$ from $a^\dagger a$

* When the vacuum has spin Ω_j , $j > 0$

it belongs to a $(2j+1)$ -dim rep of the little group $SU(2)$

→ This leads to a multiplet w/

Ω

$$\bar{j}-1/2 \quad \bar{j} \quad \bar{j}+1/2$$

$$a_1 \Omega \quad a_2 \Omega$$

$$\downarrow \quad \nwarrow$$

Super fields:

$$G(x, \theta, \bar{\theta}) := e^{-x^m P_m + \epsilon Q + \bar{\epsilon} \bar{Q}}$$

Hausdorff: $e^A e^B = e^{A+B} + \frac{1}{2}[A, B] + \dots$

$$\Rightarrow G(0, \xi, \bar{\xi}) G(x^m, \epsilon, \bar{\epsilon}) =$$

$$\exp(-x^m P_m + (\xi + \epsilon) Q + (\bar{\xi} + \bar{\epsilon}) \bar{Q})$$

$$+ [\xi Q + \bar{\xi} \bar{Q}, -x^m P_m + \epsilon Q + \bar{\epsilon} \bar{Q}]$$

only cross terms remain

$$[\xi Q, \bar{\epsilon} \bar{Q}] = \bar{\epsilon}^{\dot{\alpha}} \xi^{\alpha} \{Q_{\alpha}, \bar{Q}_{\dot{\alpha}}\}$$

$$[\bar{\xi} \bar{Q}, \epsilon Q] = \epsilon^{\alpha} \bar{\xi}^{\dot{\alpha}} \{Q_{\alpha}, \bar{Q}_{\dot{\alpha}}\} =$$

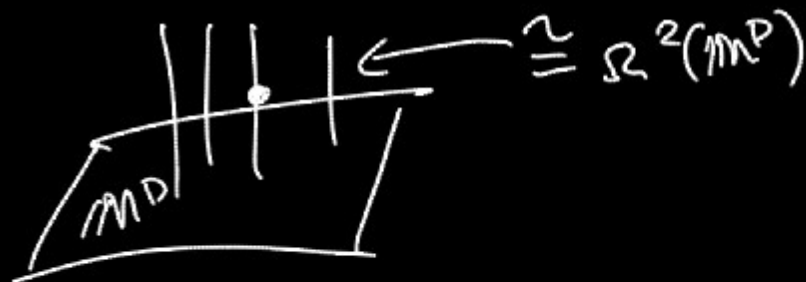
extra factors of

$$\therefore \frac{1}{2} [\cdot, \cdot] = \epsilon \sigma^m_{\dot{\alpha}\alpha} \bar{\xi} P_m - \bar{\epsilon} \sigma^m_{\alpha\dot{\alpha}} \xi P_m$$

$$\therefore G(0, \xi, \bar{\xi}) G(x^m, \epsilon, \bar{\epsilon})$$

$$= G(x^m + \epsilon \sigma^m_{\dot{\alpha}\alpha} \bar{\xi} - \bar{\xi} \sigma^m_{\alpha\dot{\alpha}} \epsilon, \theta + \xi, \bar{\theta} + \bar{\xi})$$

This motion:



The group action $g(\xi, \bar{\xi}) : (x^m, \epsilon, \bar{\epsilon})$

$$\mapsto (x^m + \epsilon \sigma^m_{\dot{\alpha}\alpha} \bar{\xi} - \bar{\xi} \sigma^m_{\alpha\dot{\alpha}} \epsilon, \theta + \xi, \bar{\theta} + \bar{\xi})$$

moves ϵ & $\bar{\epsilon}$ further along $M^{D/2}$ coords as well as moving M^D coord. x^m

- Parallel transport along $T_p M$ is captured by $D_m = \partial_m = \partial(b_m)$

However, note, here we are moving along θ and $\bar{\theta}$ so $\partial_m \rightarrow \frac{\partial}{\partial \theta}$ and $\frac{\partial}{\partial \bar{\theta}}$

$$e^{\xi Q + \bar{\xi} \bar{Q}} \gamma^m = \gamma^m + \xi Q \gamma^m + \bar{\xi} \bar{Q} \gamma^m$$

$$Q \gamma^m = -\sigma_{\alpha\dot{\alpha}}^m \bar{\theta}^{\dot{\alpha}}$$

$$\bar{Q} \gamma^m = \theta^\alpha \sigma_{\alpha\dot{\alpha}}^m$$

$$e^{\xi Q + \bar{\xi} \bar{Q}} \theta = \theta + \xi \cdot 1$$

$$\bar{Q} \theta = 0$$

$$\bar{Q} \theta = 1 \Rightarrow \bar{Q} \sim \frac{\partial}{\partial \theta}$$

$$Q \sim \frac{\partial}{\partial \bar{\theta}_{\dot{\alpha}}}$$

but also

$$Q_\alpha \gamma^m = -\sigma_{\alpha\dot{\alpha}}^m \bar{\theta}^{\dot{\alpha}}$$

$$Q_p \gamma^m = \delta_p^m$$

$$Q_\alpha \sim -\sigma_{\alpha\dot{\alpha}}^m \bar{\theta}^{\dot{\alpha}} \partial_m$$

$$\Rightarrow Q_\alpha = \frac{\partial}{\partial \theta^\alpha} - \sigma_{\alpha\dot{\alpha}}^m \bar{\theta}^{\dot{\alpha}} \partial_m$$

$$\bar{Q}^{\dot{\alpha}} = \frac{\partial}{\partial \bar{\theta}_{\dot{\alpha}}} - \theta^\alpha \sigma_{\alpha\dot{\alpha}}^m \epsilon^{\dot{\beta}\alpha} \partial_m$$

$$[A^\alpha, A^\beta] = 0 \quad [A^\alpha, B^\beta] = 0 \quad [B^\alpha, B^\beta] = 0$$

Recall $Sp(2n, F) := \{ M \in M_{2n \times 2n}(F) \mid M^T \Omega M = \Omega \}$
 and $\Omega = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$

Here: Here, suppose $\Omega = \begin{pmatrix} \Omega_2 & \Omega_N \\ \Omega_N & \Omega_2 \end{pmatrix}$
 $\begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} \Omega_2 & \Omega_N \\ \Omega_N & \Omega_2 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} \Omega_2 & \Omega_N \\ \Omega_N & \Omega_2 \end{pmatrix}$

$$\begin{pmatrix} A^T & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \Omega_2 & \Omega_N \\ \Omega_N & \Omega_N \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} = \begin{pmatrix} A^T & 0 \\ 0 & D^T \end{pmatrix} \begin{pmatrix} \Omega_2 A & \Omega_N D \\ \Omega_N A & \Omega_N D \end{pmatrix} \\ = \begin{pmatrix} A^T \Omega_2 A & A^T \Omega_N D \\ D^T \Omega_N A & D^T \Omega_N D \end{pmatrix} = \begin{pmatrix} \Omega_2 & \Omega_N \\ \Omega_N & \Omega_N \end{pmatrix}$$

$$\boxed{\begin{matrix} A^T \Omega_2 A = \Omega_2 \\ D^T \Omega_N D = \Omega_N \end{matrix}}$$

$$\begin{pmatrix} A^T & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} \Omega_2 & 0 \\ 0 & \Omega_N \end{pmatrix} \begin{pmatrix} A^T & 0 \\ 0 & D \end{pmatrix} = \begin{pmatrix} \Omega_2 A & 0 \\ 0 & \Omega_N D \end{pmatrix} \\ A^T \Omega_2 A = \Omega_2 \\ D^T \Omega_N D = \Omega_N$$

$$\begin{pmatrix} 0 & C^T \\ B^T & 0 \end{pmatrix} \begin{pmatrix} \Omega_2 & 0 \\ 0 & \Omega_N \end{pmatrix} \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix} = \begin{pmatrix} 0 & C^T \\ B^T & 0 \end{pmatrix} \begin{pmatrix} 0 & \Omega_2 B \\ \Omega_N C & 0 \end{pmatrix} \\ = \begin{pmatrix} C^T \Omega_N C & B^T \Omega_2 B \\ 0 & 0 \end{pmatrix}$$

$$\forall A^T \Omega_{(1,p)} A = \Omega_{(1,p)}$$

$$\Omega^T_{(N)} = \Omega_{(N)}$$

ohhhh --- the given expansion is for the Lie Algebra

$$A = e^{\tilde{a}} \Rightarrow (1 + \tilde{a})^T \Omega (1 + \tilde{a}) = \Omega$$

$$\tilde{a}^T \Omega + \Omega \tilde{a} = 0$$

For

$$C = \Omega_{(N)} B^T \Omega_{(2p)}$$

$$\Rightarrow \mathbb{L}_0 \cong \mathfrak{sp}(2p) \oplus \mathcal{O}(N)$$

$$Q \in \mathcal{Osp}(2p|N)$$

$$Q = \begin{array}{c} g^a t_a \\ \uparrow \\ \mathbb{L}_0 \end{array} + \begin{array}{c} g^L t_L \\ \uparrow \\ \mathbb{L}_1 \end{array} \quad \left| \begin{array}{l} g^a \in \mathbb{R} \\ g^L \in \mathbb{L}^2(\mathbb{R}) \end{array} \right.$$

N -extended susy alg. in Minkowski space: $\overline{\mathcal{Osp}(2p|N)}$

$$A \mapsto P_\mu, M_{\mu\nu} \quad D \mapsto Z^{\pm 5}$$

$$B, C \mapsto Q_\pm, \bar{Q}_\pm$$

$$\mathcal{M}^{4|1} := \frac{\mathcal{Osp}(4|1)}{SO(1,3)} \quad (N=1 \text{ superspace})$$

• A point in superspace gets identified w/ the coset rep. for super-translations

$$(x^\mu, \theta_\alpha, \bar{\theta}_{\dot{\alpha}}) \mapsto \exp(x^\mu P_\mu) \exp(\theta Q + \bar{\theta} \bar{Q})$$

* Given a scalar field $\phi(x)$

$$\hat{T}_1 = -g^{\mu\nu} P_\mu \hat{T}_2 = -g^{\mu\nu} P_\nu \quad (\text{translation})$$

$\hat{\phi}$ is also a Hilbert vector, so $\exists$ BP.

on $\mathcal{H}$ s.t.

$$\phi \mapsto e^{i\hat{p}_n} \hat{p}_n \phi =: \phi(x^n - a^n)$$

here $\hat{p}_n = \partial_n$

$$\Rightarrow (1 - a^n \hat{p}_n) \hat{\phi} (1 + a^n \hat{p}_n) = (1 - a^n \hat{p}_n) \hat{\phi}$$

$$(\hat{\phi} - a^n \hat{p}_n \hat{\phi})(1 + a^n \hat{p}_n) =$$

$$\hat{\phi} - a^n \hat{p}_n \hat{\phi} + \hat{\phi} a^n \hat{p}_n = \hat{\phi} - a^n \hat{p}_n \hat{\phi}$$

$$\therefore [\hat{\phi}, a^n \hat{p}_n] = -a^n \hat{p}_n \hat{\phi}$$

A generic Superfield Φ is of the form

$$\begin{aligned} \Phi(x, \theta, \bar{\theta}) = & f(x) + \theta \psi(x) + \bar{\theta} \bar{\chi}(x) + \theta \theta (m(x)) \\ & + \bar{\theta} \bar{\theta} n(x) + \theta \sigma \bar{\theta} V_n(\theta) + \theta \theta \bar{\theta} \bar{\chi}(x) \\ & + \bar{\theta} \bar{\theta} (\theta \rho(x)) + (\theta \theta) (\bar{\theta} \bar{\theta}) d(x) \end{aligned}$$

$$\begin{aligned} \Phi(x + \delta x, \theta + \delta \theta, \bar{\theta} + \delta \bar{\theta}) &= e^{-i(\epsilon Q + \bar{\epsilon} \bar{Q})} \Phi(x, \theta, \bar{\theta}) e^{i(\epsilon Q + \bar{\epsilon} \bar{Q})} \\ \delta_{\epsilon, \bar{\epsilon}} \Phi(x, \theta, \bar{\theta}) &:= \Phi(x + \delta x, \theta + \delta \theta, \bar{\theta} + \delta \bar{\theta}) - \Phi(x, \theta, \bar{\theta}) \end{aligned}$$

Note

$$\Phi(x, \theta, \bar{\theta}) = e^{-i(x^n \hat{p}_n + \theta Q + \bar{\theta} \bar{Q})} \Phi(0, 0, 0) e^{i(x^n \hat{p}_n + \theta Q + \bar{\theta} \bar{Q})}$$

$$\therefore \Phi(x + \delta x, \theta + \delta \theta, \bar{\theta} + \delta \bar{\theta}) = e^{-i(\epsilon Q + \bar{\epsilon} \bar{Q})} e^{-i(x^n \hat{p}_n + \theta Q + \bar{\theta} \bar{Q})} \Phi(0, 0, 0) e^{i(x^n \hat{p}_n + \theta Q + \bar{\theta} \bar{Q})} e^{i(\epsilon Q + \bar{\epsilon} \bar{Q})}$$

LHS: $\exp(i\epsilon Q + \bar{\epsilon} \bar{Q}) \exp(i(x^n \hat{p}_n + \theta Q + \bar{\theta} \bar{Q}))$
 $e^A e^B = e^C, \quad C = A + B + \frac{1}{2}[A, B] + \frac{1}{4}[A, [A, B]] - \frac{1}{2} \dots$

$$\exp(i(\epsilon + \theta)Q + i(\bar{\epsilon} + \bar{\theta})\bar{Q} + iX^\mu P_\mu + \frac{1}{2}L;)$$

$$\cdot [EQ + \bar{E}\bar{Q}, \bar{\theta}\bar{Q} + \theta Q]$$

$$\begin{aligned} &= -\frac{1}{2} \{ [EQ, \bar{\theta}\bar{Q}] + [\bar{E}\bar{Q}, \theta Q] \} \\ &= -\frac{1}{2} \{ \bar{\theta}^\alpha \epsilon^\alpha 2\sigma^\mu_{\alpha\dot{\alpha}} P_\mu - 2\bar{E}^{\dot{\alpha}} \theta^\alpha \sigma^\mu_{\alpha\dot{\alpha}} P_\mu \} \\ &= \epsilon \sigma^\mu \bar{\theta} P_\mu - \theta \sigma^\mu \bar{E} P_\mu \\ &= i(i\theta \sigma^\mu \bar{E} P_\mu - \epsilon \sigma^\mu \bar{\theta} P_\mu) \end{aligned}$$

$\therefore$

$$\begin{aligned} &\exp(i(\epsilon Q + \bar{E}\bar{Q})) \exp(i(X^\mu P_\mu + \theta Q + \bar{\theta}\bar{Q})) \\ &= \exp[i(X^\mu + i\theta \sigma^\mu \bar{E} + \epsilon \sigma^\mu \bar{\theta}) P_\mu + i(\theta + \epsilon)Q \\ &\quad + i(\bar{\theta} + \bar{E})\bar{Q}] \end{aligned}$$

$$\begin{cases} \delta x = i\theta \sigma^\mu \bar{E} - i\epsilon \sigma^\mu \bar{\theta} \\ \delta \theta^\alpha = \epsilon^\alpha \\ \delta \bar{\theta}^{\dot{\alpha}} = \bar{E}^{\dot{\alpha}} \end{cases}$$

$\Rightarrow$ Supp transformations translate M^4 coord into $M^{(4|2)}$

$$\text{Now, } \delta_{\epsilon, \bar{E}} \Phi = \Phi(x + \delta x, \theta + \delta \theta, \bar{E} + \delta \bar{E}) - \Phi$$

$$\begin{aligned} \Phi(x + \delta x, \theta + \delta \theta, \bar{E} + \delta \bar{E}) &= \Phi + \delta x^\mu \partial_\mu \Phi \\ &\quad + \delta \theta^\alpha \partial_\alpha \Phi + \delta \bar{E}^{\dot{\alpha}} \partial_{\dot{\alpha}} \Phi + \mathcal{O}((x, \theta, \bar{E})^2) \end{aligned}$$

$$\delta_{\epsilon, \bar{E}} \Phi = [\epsilon^\alpha \partial_\alpha + \bar{E}^{\dot{\alpha}} \partial_{\dot{\alpha}} + i(\theta \sigma^\mu \bar{E} - \epsilon \sigma^\mu \bar{\theta}) \partial_\mu] \Phi$$

However,

$$\Phi(x+\delta x, \dots) = e^{-i\epsilon Q} \Phi e^{i\epsilon Q}$$

$$= (1 - i(\epsilon Q)) \Phi (1 + i\epsilon Q)$$

[Note $\bar{\epsilon}_j \bar{Q}^{\dot{\alpha}} = \epsilon^{\dot{\alpha} \beta} \bar{\epsilon}_j \bar{Q}_{\dot{\beta}} = -\epsilon^{\dot{\alpha} \beta} \bar{Q}_{\dot{\beta}} \bar{\epsilon}_j = -\bar{Q}^{\dot{\alpha}} \bar{\epsilon}_j$

$$(1 - i\epsilon Q - i\epsilon Q) \Phi (1 + i\epsilon Q + i\epsilon Q)$$

$$(1 - i\epsilon Q - i\epsilon Q) \Phi (1 + i\epsilon Q + i\epsilon Q)$$

$$= \Phi - i\epsilon Q \Phi - i\epsilon Q \Phi + i\Phi \epsilon Q + i\Phi \epsilon Q$$

$$= \Phi + i[\Phi, \epsilon Q] + i[\Phi, \epsilon Q]$$

$$= \Phi + i\epsilon^{\alpha} [\Phi, Q_{\alpha}] - i\bar{\epsilon}^{\dot{\alpha}} [\Phi, \bar{Q}_{\dot{\alpha}}]$$

$$\delta \epsilon, \bar{\epsilon} = i\epsilon^{\alpha} [\Phi, Q_{\alpha}] - i\bar{\epsilon}^{\dot{\alpha}} [\Phi, \bar{Q}_{\dot{\alpha}}]$$

Defining $[\Phi, Q_{\alpha}] \equiv \hat{Q}_{\alpha} \Phi$
 $[\Phi, \bar{Q}_{\dot{\alpha}}] \equiv \hat{\bar{Q}}_{\dot{\alpha}} \Phi$

$\Rightarrow$ Can easily read off $\hat{Q}_{\alpha}$ & $\hat{\bar{Q}}_{\dot{\alpha}}$

$$i\hat{Q}_{\alpha} = \partial_{\alpha} - i\sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_{\mu}$$

$$i\hat{\bar{Q}}_{\dot{\alpha}} = \bar{\partial}_{\dot{\alpha}} + i\sigma^{\mu}_{\alpha\dot{\alpha}} \theta^{\alpha} \partial_{\mu}$$

$$d^2\theta := \frac{1}{2} d\theta^1 d\theta^2$$

$$d^2\bar{\theta} := \frac{1}{2} d\bar{\theta}^{\dot{1}} d\bar{\theta}^{\dot{2}}$$

$$\int d^2\theta \theta\theta = 1$$

Proof: $\int \frac{1}{2} d\theta^1 d\theta^2 (\epsilon_{\alpha\beta} \theta^{\alpha} \theta^{\beta})$

$$= \frac{1}{2} \int d\theta^1 d\theta^2 (-\theta^1 \theta^2 + \theta^2 \theta^1)$$

$$= 1 \quad \square$$

$$\Rightarrow \boxed{\int d^2\theta = \frac{1}{4} \epsilon^{\alpha\beta} \partial_\alpha \partial_\beta}$$

$$\int d^4x d^2\theta d^2\bar{\theta} \Phi(x, \theta, \bar{\theta}) \quad (\text{Susy invariant})$$

proof:

$$\int d\theta \theta = 1$$

$$\theta \rightarrow \theta + \xi \Rightarrow \int d(\theta + \xi) (\theta + \xi) = 1$$

So it is translationally invariant.

$$\delta \epsilon, \bar{\epsilon} \int d^4x d^2\theta d^2\bar{\theta} \Phi$$

$$= \int d^4x d^2\theta d^2\bar{\theta} \left(\epsilon^\alpha \partial_\alpha \Phi + \bar{\epsilon}^{\dot{\alpha}} \partial_{\dot{\alpha}} \Phi + \partial_m (\dots) \right)$$

→ Do not have enough θ 's

→ Total derivative.

$$\{iQ_\alpha, i\bar{Q}_{\dot{\alpha}}\} = \{\partial_\alpha - i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_m, -\partial_{\dot{\alpha}} + i\theta^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \partial_m\}$$

$$= \{\cancel{\partial_\alpha}, \cancel{\partial_{\dot{\alpha}}}\} + \{\partial_\alpha, i\theta^\beta \sigma^\mu_{\beta\dot{\alpha}} \partial_m\} - \{i\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_m, \partial_{\dot{\alpha}}\} + \{\sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \partial_m, \cancel{\sigma^\mu_{\alpha\dot{\alpha}} \theta^\alpha \partial_m}\}$$

$$= i\sigma^\mu_{\alpha\dot{\alpha}} \partial_m + i\sigma^\mu_{\alpha\dot{\alpha}} \partial_m = 2\sigma^\mu_{\alpha\dot{\alpha}} P_m$$

$$\therefore \{Q_\alpha, \bar{Q}_{\dot{\alpha}}\} = 2\sigma^\mu_{\alpha\dot{\alpha}} P_m$$

on $\mathcal{N}^{(4|2)}$

Invariance: For $\Phi(x^\mu, \theta_\alpha, \bar{\theta}_{\dot{\alpha}})$

$$\delta_{\epsilon, \bar{\epsilon}} \Phi = i(\epsilon Q + \bar{\epsilon} \bar{Q}) \Phi$$

General Φ

$$\delta \Phi = \delta \Phi + \theta \delta \psi + \bar{\theta} \delta \bar{\chi}(x) + \dots$$

$$i \epsilon^\alpha (-i \partial_\alpha - \bar{\theta}^{\dot{\alpha}} \sigma^\mu_{\alpha \dot{\alpha}} \partial_\mu) + i \bar{\epsilon}^{\dot{\alpha}} (\partial_{\dot{\alpha}} - \theta^\beta \sigma^\mu_{\beta \dot{\alpha}} \partial_\mu) \\ (\Phi + \theta \psi + \bar{\theta} \chi + \theta \theta M(x) + \bar{\theta} \bar{\theta} \Lambda + \theta \sigma^\mu \bar{\theta} V_\mu \\ + \theta \theta \bar{\theta} \bar{\chi} + \bar{\theta} \bar{\theta} \theta \psi + \theta \theta \bar{\theta} \bar{\theta} D(x))$$

$$\Rightarrow \epsilon^\alpha i \cdot \sigma^\mu_{\alpha \dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_\mu \psi + i \bar{\epsilon}^{\dot{\alpha}} \theta^\beta \sigma^\mu_{\beta \dot{\alpha}} \partial_\mu \chi \\ \left[\begin{aligned} \epsilon^\alpha \psi_\alpha &= \epsilon^\alpha \bar{\theta}^{\dot{\alpha}} \sigma^\mu_{\alpha \dot{\alpha}} \partial_\mu \theta^\beta \psi_\beta \\ &- i \bar{\epsilon}^{\dot{\alpha}} \theta^\beta \sigma^\mu_{\beta \dot{\alpha}} \partial_\mu (\theta^\gamma \psi_\gamma) \\ &- i \epsilon^\alpha \bar{\theta}^{\dot{\alpha}} \sigma^\mu_{\alpha \dot{\alpha}} \theta^\beta \partial_\mu \psi_\beta \\ &- i \delta^\alpha_\gamma \epsilon^\gamma \bar{\theta}^{\dot{\alpha}} \sigma^\mu_{\alpha \dot{\alpha}} \delta^\beta_\epsilon \theta^\epsilon \partial_\mu \psi_\beta \end{aligned} \right. \\ = -i$$

$$\rightarrow \text{Tr}(\sigma^\mu \bar{\sigma}^\nu) = 2\eta^{\mu\nu} \\ \epsilon^\alpha (\bar{\sigma}^\mu)_{\dot{\alpha} \beta} (\sigma^\mu)^\beta_{\dot{\gamma}} = \delta^{\dot{\alpha}}_{\dot{\gamma}} \\ \theta^\gamma \theta_\beta = \frac{1}{2} \delta^\gamma_\beta (\theta \theta)$$

$$\boxed{\sigma^\mu_{\alpha \dot{\alpha}} (\bar{\sigma}^\mu)^{\dot{\beta} \beta} = 2 \delta^\beta_\alpha \delta^{\dot{\beta}}_{\dot{\alpha}}}$$

$$= -i \delta^\alpha_\gamma \epsilon^\gamma \bar{\epsilon}^{\dot{\alpha}} \delta^{\dot{\beta}}_{\dot{\alpha}} \sigma^\mu_{\alpha \dot{\alpha}} \theta^\beta \partial_\mu \psi_\beta \\ = -i (2 \sigma^\mu_{\gamma \dot{\alpha}} \sigma^\mu_{\alpha \dot{\beta}}) \epsilon^\gamma \sigma^\nu_{\alpha \dot{\alpha}} \theta^\beta \partial_\nu \psi_\beta \\ = -2i \epsilon^\gamma (\sigma^\mu)_{\gamma \dot{\alpha}} (\sigma^\mu)_{\alpha \dot{\beta}} (\sigma^\nu)_{\alpha \dot{\alpha}} \bar{\theta}^{\dot{\beta}} \theta^\beta \partial_\nu \psi_\beta \\ = -2i \epsilon^\gamma (\sigma^\mu)_{\gamma \dot{\alpha}} (\sigma^\mu \sigma^\nu)_{\dot{\alpha} \beta} \bar{\theta}^{\dot{\beta}} \theta^\beta \partial_\nu \psi_\beta \\ = +2i (\epsilon \sigma^\mu \bar{\theta}) (\sigma^\mu \sigma^\nu)_{\dot{\alpha} \beta} \partial_\nu \psi_\beta$$

Covariant Derivatives:

$$\begin{cases} D_\alpha := \partial_\alpha + i \sigma_{\alpha\dot{\alpha}}^n \bar{\theta}^{\dot{\alpha}} \partial_n \\ D_{\dot{\alpha}} := \bar{\partial}_{\dot{\alpha}} + i \end{cases}$$

$\Rightarrow \partial_m S$ is a superfield but $\partial_\alpha S$ is not.

Proof:

$$\delta(\partial_\alpha S) = \partial_\alpha(\delta S) = i \partial_\alpha [(\epsilon Q + \bar{\epsilon} \bar{Q}) S]$$

however

$$[\partial_\alpha, \bar{\epsilon} \bar{Q}_{\dot{\alpha}}] \neq 0$$

$$\partial_\alpha \bar{\epsilon}^{\dot{\beta}} (i \partial_{\dot{\alpha}} + \theta^{\beta} \sigma_{\beta\dot{\beta}}^n \partial_n) = () \partial_\alpha$$

$$= \epsilon^{\beta} (-i \delta_{\alpha}^{\beta} \sigma_{\beta\dot{\beta}}^n \partial_n) \neq 0$$

$\therefore$ For $\partial_\alpha S$ to transform covariantly under supersymmetry we define

D_α and $D_{\dot{\alpha}}$ as above.

s.t

$$[D_\alpha, \epsilon Q + \bar{\epsilon} \bar{Q}] = 0$$

* Given $(D_\alpha, D_{\dot{\alpha}})$ we define

(i) Chiral Superfield $\Phi^i(x^n, \theta_\alpha, \bar{\theta}_{\dot{\alpha}})$

$$\boxed{D_{\dot{\alpha}} \Phi = 0}$$

(ii) Anti-chiral: $\boxed{D_\alpha \bar{\Phi} = 0}$

(iii) Vector/Real: $\boxed{V^\dagger := V}$

(iv) Linear superfield L

$$\boxed{(DDL) \wedge (L = 1^+)}$$

- For a chiral superfield ($\bar{D}_{\dot{\alpha}} \Phi = 0$) we can define coordinates in $M^{(4|2)}$

$$y^m = x^m + i \theta \sigma^m_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}}$$

$$\bar{y}^{\dot{m}} := x^m - i \theta \sigma^m_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}}$$

Claim: (a) $\bar{D}_{\dot{\alpha}} y^m = 0$

(b) $\bar{D}_{\dot{\alpha}} \theta_{\alpha} = 0$

Proof:

$$\begin{aligned} (a) & (\bar{D}_{\dot{\alpha}} + i \theta^{\beta} \sigma^{\mu}_{\beta\dot{\alpha}} \partial_{\mu}) (x^m + i \theta^{\gamma} \sigma^m_{\gamma\dot{\gamma}} \bar{\theta}^{\dot{\gamma}}) \\ &= \bar{D}_{\dot{\alpha}} (i \theta^{\gamma} \sigma^m_{\gamma\dot{\gamma}} \bar{\theta}^{\dot{\gamma}}) + i \theta^{\beta} \sigma^{\mu}_{\beta\dot{\alpha}} \partial_{\mu} x^m \\ &= -i \theta^{\gamma} \sigma^m_{\gamma\dot{\gamma}} \delta^{\dot{\gamma}}_{\dot{\alpha}} + i \theta^{\beta} \sigma^{\mu}_{\beta\dot{\alpha}} \delta^m_{\mu} \\ &= -i \theta^{\gamma} \sigma^m_{\gamma\dot{\alpha}} + i \theta^{\gamma} \sigma^m_{\gamma\dot{\alpha}} = 0 \quad \checkmark \end{aligned}$$

(b) $\bar{D}_{\dot{\alpha}} \theta_{\alpha} = 0$ (trivially).

Similarly, we have $D_{\alpha} \bar{\theta}_{\dot{\beta}} = D_{\alpha} \bar{y}^{\dot{m}} = 0$

Claim: The chiral-superfield can be written in terms of $(y^m; \theta_{\alpha})$

Proof: (i) First show that under y^m the (super) covariant derivatives are:

$$D_{\alpha}(x, \theta, \bar{\theta}) = \frac{\partial}{\partial \theta^{\alpha}} + i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_{\mu}$$

$$\bar{D}_{\dot{\alpha}}(x, \theta, \bar{\theta}) = -\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} - i \theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu}$$

$$\Rightarrow \frac{\partial}{\partial x^{\nu}} = \frac{\partial y^m}{\partial x^{\nu}} \frac{\partial}{\partial y^m} = \delta^m_{\nu} \frac{\partial}{\partial y^m} = \frac{\partial}{\partial y^{\nu}}$$

$$\begin{aligned} \frac{\partial}{\partial \theta^{\alpha}} &= \frac{\partial \theta^{\beta}}{\partial \theta^{\alpha}} \frac{\partial}{\partial \theta^{\beta}} + \frac{\partial \bar{\theta}^{\dot{\beta}}}{\partial \theta^{\alpha}} \frac{\partial}{\partial \bar{\theta}^{\dot{\beta}}} + \frac{\partial y^m}{\partial \theta^{\alpha}} \frac{\partial}{\partial y^m} \\ &= \delta^{\beta}_{\alpha} \frac{\partial}{\partial \theta^{\beta}} + i \delta^{\mu}_{\alpha} \sigma^m_{\mu\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \left(\frac{\partial}{\partial y^m} \right) \end{aligned}$$

where $\theta'^\alpha := \theta^\alpha$

$$\frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} = \frac{\partial \bar{\theta}^{\dot{\beta}}}{\partial \bar{\theta}^{\dot{\alpha}}} \frac{\partial}{\partial \bar{\theta}^{\dot{\beta}}} + \frac{\partial \theta^\beta}{\partial \bar{\theta}^{\dot{\alpha}}} \frac{\partial}{\partial \theta^\beta} + \frac{\partial y^\mu}{\partial \theta^\alpha} \frac{\partial}{\partial y^\mu}$$

$$= \frac{\partial}{\partial \bar{\theta}^{\dot{\alpha}}} - i \theta^\alpha \sigma_\alpha^{\mu \dot{\alpha}} \frac{\partial}{\partial y^\mu}$$

$$\begin{aligned} \therefore D_\alpha(x, \theta, \bar{\theta}) &= \partial_\alpha + i \sigma_{\alpha \dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} \\ &= \left[\partial_\alpha + i \sigma_{\alpha \dot{\alpha}}^\mu \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} \right] \\ &\quad + i \theta^\beta \sigma_{\beta \dot{\alpha}}^\mu \left(\frac{\partial}{\partial y^\mu} \right) \\ &= \partial_\alpha + 2i \theta^\beta (\sigma^\mu)_{\beta \dot{\alpha}} \left(\frac{\partial}{\partial y^\mu} \right) =: D_\alpha^{(2)}(y, \theta) \end{aligned}$$

and, for $\bar{D}_{\dot{\alpha}}$:

$$\begin{aligned} \bar{D}_{\dot{\alpha}}(x, \theta, \bar{\theta}) &:= \bar{\partial}_{\dot{\alpha}} + i \theta^\beta \sigma_{\beta \dot{\alpha}}^\mu \frac{\partial}{\partial y^\mu} \\ &= \left[\bar{\partial}_{\dot{\alpha}} - i \theta'^\beta \sigma_{\beta \dot{\alpha}}^\mu \frac{\partial}{\partial y^\mu} \right] + i \theta^\beta \sigma_{\beta \dot{\alpha}}^\mu \left(\frac{\partial}{\partial y^\mu} \right) \\ &= \bar{\partial}_{\dot{\alpha}} =: \bar{D}_{\dot{\alpha}}^{(1)}(y, \theta', \bar{\theta}') \end{aligned}$$

Therefore, the constraint on a chiral Superfield Φ :

$$\bar{D}_{\dot{\alpha}} \Phi = 0$$

Simply becomes

$$\partial_{\dot{\alpha}} \Phi^{(1)}(y, \theta', \bar{\theta}') = 0$$

where

$$\Phi^{(1)}(y, \theta', \bar{\theta}') := \Phi(y^\mu - i \theta' \sigma^\mu \bar{\theta}', \theta', \bar{\theta}')$$

$\Leftrightarrow \Phi^{(1)}$ is, in superspace, basically "holomorphic" with respect to $\bar{\theta}^{\dot{\alpha}}$

$$\therefore \Phi^{(1)}(y, \bar{\theta}', \theta') = \Phi^{(1)}(y^\mu, \theta')$$

so $\Phi^{(1)}$ has a power-series expansion in θ as

$$\boxed{\mathcal{L}^{(1)}(y^\mu, \theta_\alpha) := \phi(y) + \sqrt{2} \theta^\alpha \psi_\alpha(y) - \theta \theta F(y)} \quad \square$$

* $\psi_\alpha(y)$: Left-handed Chiral Spinor

$\in \text{SUC}(2) \supseteq \text{rep } \square$

of $\text{SO}(1,3) \cong \text{SUC}(2) \times \overline{\text{SUC}(2)}$

ϕ, F : Complex ($\mathbb{C}$) scalar-valued fields
 $\underline{1}$ of $\text{SO}(1,3)$.

The Full expansion in $M^{(1,2)}$ of $(x, \theta, \bar{\theta})$
 is given by $y^\mu = x^\mu + i \theta \sigma^\mu \bar{\theta}$

$$\Rightarrow \mathcal{L}(y, \theta) = \phi(y) + \sqrt{2} \theta^\alpha \psi_\alpha(y) - \theta \theta F(y)$$

$$= \phi(x + i \theta \sigma^\mu \bar{\theta}) + \sqrt{2} \theta^\alpha \psi_\alpha(x + i \theta \sigma^\mu \bar{\theta})$$

$$- \theta \theta F(x + i \theta \sigma^\mu \bar{\theta})$$

$$\left[= \phi(x) + i \theta \sigma^\mu \bar{\theta} \partial_\mu \phi(x) + \sqrt{2} \theta^\alpha \psi_\alpha(x) \right.$$

$$\left. + \sqrt{2} i \theta^\alpha \theta^\gamma (\sigma^\mu_{\gamma\delta}) \bar{\theta}^\delta \partial_\mu \psi_\alpha(x) \right.$$

$$\left. - \theta \theta F(x) - 0 + (i \theta \sigma^\mu \bar{\theta} i \theta \sigma^\nu \bar{\theta}) \partial_\mu \partial_\nu \phi \right.$$

$$= \phi(x) + i \theta \sigma^\mu \bar{\theta} \partial_\mu \phi(x) + \sqrt{2} \theta^\alpha \psi_\alpha(x)$$

$$\rightarrow i \theta^\alpha \theta^\gamma (\sigma^\mu_{\gamma\delta}) \bar{\theta}^\delta \partial_\mu \psi_\alpha(x)$$

$$\left[\text{Claim: } -\frac{1}{2} \varepsilon^{\alpha\gamma} (\theta \theta) = \theta^\alpha \theta^\gamma \right.$$

$$\text{Proof: } -\frac{1}{2} \varepsilon^{\alpha\gamma} (\theta^\beta \theta_\beta) = -\frac{1}{2} \varepsilon^{\alpha\gamma} \varepsilon_{\beta\delta} \theta^\beta \theta^\delta$$

$$\left[\varepsilon^{\alpha\gamma} \varepsilon_{\beta\delta} = -\varepsilon^{\alpha\delta} \varepsilon_{\beta\gamma} \right]$$

$$\begin{aligned} &= -\frac{1}{2} (-2 \delta_{\beta}^{\alpha} \delta_{\delta}^{\gamma}) \theta^{\beta} \theta^{\delta} = \delta_{\beta}^{\alpha} \delta_{\delta}^{\gamma} \theta^{\beta} \theta^{\delta} \\ &= \frac{1}{2} (\theta^{\alpha} \theta^{\gamma} - \theta^{\gamma} \theta^{\alpha}) = \underline{\theta^{\alpha} \theta^{\gamma}} \quad \square \end{aligned}$$

$$\begin{aligned} &= +i \left(-\frac{1}{\sqrt{2}} \epsilon^{\alpha\gamma} (\theta\theta) \sigma^{\mu}_{\gamma\dot{\gamma}} \bar{\theta}^{\dot{\gamma}} \partial_{\mu} \phi_{\alpha}(x) \right) \\ &= -\frac{i}{\sqrt{2}} (\theta\theta) \sigma^{\mu}_{\gamma\dot{\gamma}} \partial_{\mu} \phi^{\gamma}(x) \bar{\theta}^{\dot{\gamma}} \\ &= -\frac{i}{\sqrt{2}} (\partial_{\mu} \phi \sigma^{\mu} \bar{\theta}) (\theta\theta) \quad \checkmark \end{aligned}$$

$$-\frac{1}{2} (\theta\sigma_{\mu}\bar{\theta}) (\theta\sigma^{\nu}\bar{\theta}) \partial^{\mu} \partial^{\nu} \phi(x)$$

$$\text{Claim: } (\theta\sigma^{\mu}\bar{\theta}) (\theta\sigma^{\nu}\bar{\theta}) = \frac{1}{2} \eta^{\mu\nu} (\theta\theta) (\bar{\theta}\bar{\theta})$$

$$\Rightarrow -\frac{1}{4} (\theta\theta) (\bar{\theta}\bar{\theta}) \square \phi(x)$$

$\therefore$

$$\begin{aligned} \Phi(x, \theta, \bar{\theta}) &= \phi(x) + \sqrt{2} \theta \psi + i \theta \sigma^{\mu} \bar{\theta} \partial_{\mu} \phi(x) - \theta \theta F \\ &\quad - \frac{i}{\sqrt{2}} \theta \theta \partial_{\mu} \phi \sigma^{\mu} \bar{\theta} - \frac{1}{4} (\theta\theta) (\bar{\theta}\bar{\theta}) \square \phi(x) \end{aligned}$$

* This can also be seen as a translation from the $(x, \theta, 0)$ plane into all of $M^{(4|1)}$

$$\Phi(x, \theta, \bar{\theta}) = e^{i \theta \sigma^{\mu} \bar{\theta} \partial_{\mu}} \Phi(x, \theta)$$

* Anti-chiral Superfield:

$$\bar{\Phi}(x, \theta, \bar{\theta}) = \phi^{\dagger}(\bar{y}) + \sqrt{2} \bar{\theta} \bar{\psi}(\bar{y}) - \bar{\theta} \bar{\theta} F^{\dagger}(\bar{x})$$

Supersymmetry transformations:

$$\Sigma \quad \bar{\Phi} = i (\sigma^{\mu} \epsilon \bar{\theta}) \bar{\Phi}$$

$\Rightarrow$ Here in $(y^m, \theta^r, \bar{\theta}^s)$

$$Q_{\alpha}^{(1)} = -i \partial_{\alpha}$$

$$\bar{Q}_{\dot{\alpha}}^{(1)} = i \bar{\partial}_{\dot{\alpha}} + 2\theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \frac{\partial}{\partial y^{\mu}}$$

$$\begin{aligned} \therefore i \left(\epsilon^{\alpha} (-i \partial_{\alpha}) - \bar{\epsilon}^{\dot{\alpha}} \left(i \bar{\partial}_{\dot{\alpha}} + 2\theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \frac{\partial}{\partial y^{\mu}} \right) \right) \Phi^{(1)}(y, \theta) \\ \epsilon^{\alpha} \partial_{\alpha} + \bar{\epsilon}^{\dot{\alpha}} \bar{\partial}_{\dot{\alpha}} - i \bar{\epsilon}^{\dot{\alpha}} (2\theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \frac{\partial}{\partial y^{\mu}}) (\psi + \sqrt{2} \theta \psi - \theta \theta F) \\ = \sqrt{2} \epsilon \psi - i \bar{\epsilon}^{\dot{\alpha}} \theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu} \psi - \underline{i \bar{\epsilon}^{\dot{\alpha}} (2\theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu} \sqrt{2} \theta \psi)} \\ - \epsilon^{\alpha} \partial_{\alpha} (\theta \theta F) \end{aligned}$$

$$\begin{aligned} \rightarrow -i \bar{\epsilon}^{\dot{\alpha}} \theta^{\alpha} \sigma^{\mu}_{\alpha\dot{\alpha}} \sqrt{2} \theta^{\gamma} \partial_{\mu} \psi_{\gamma} \\ = -i \sqrt{2} \bar{\epsilon}^{\dot{\alpha}} \left(-\frac{1}{2} \epsilon^{\alpha\gamma} (\theta \theta) \right) \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu} \psi_{\gamma} \\ = i \sqrt{2} \bar{\epsilon}^{\dot{\alpha}} (\theta \theta) \sigma^{\mu}_{\alpha\dot{\alpha}} \partial_{\mu} \psi^{\alpha} \\ = -i \sqrt{2} (\theta \theta) (\partial_{\mu} \psi^{\alpha}) \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\epsilon}^{\dot{\alpha}} \\ = -i \sqrt{2} (\theta \theta) (\partial_{\mu} \psi \sigma^{\mu} \bar{\epsilon}) \end{aligned}$$

$$\therefore \delta_{\epsilon, \bar{\epsilon}} \Phi^{(1)} =$$

$$\begin{aligned} \sqrt{2} \epsilon \psi + i 2 \theta \sigma^{\mu} \bar{\epsilon} \partial_{\mu} \psi - i \sqrt{2} (\theta \theta) (\partial_{\mu} \psi \sigma^{\mu} \bar{\epsilon}) \\ - \underline{\epsilon^{\alpha} \partial_{\alpha} (\theta \theta) F} \end{aligned}$$

$$\begin{aligned} \rightarrow -\epsilon^{\alpha} \partial_{\alpha} (\theta^{\beta} \theta_{\beta}) F = -\epsilon^{\alpha} \epsilon_{\beta\gamma} \partial_{\alpha} (\theta^{\beta} \theta^{\gamma}) F \\ = -\epsilon^{\alpha} \epsilon_{\beta\gamma} \delta^{\beta}_{\alpha} \theta^{\gamma} F + \epsilon^{\alpha} \epsilon_{\beta\gamma} \theta^{\beta} \delta^{\gamma}_{\alpha} F \\ = \epsilon^{\alpha} (-\epsilon^{\gamma}_{\alpha} \theta^{\gamma} + \epsilon^{\beta}_{\alpha} \theta^{\beta}) F \\ = -2 \epsilon^{\alpha} \theta_{\alpha} F \end{aligned}$$

$$\therefore \delta_{\epsilon, \bar{\epsilon}} \Phi^{(1)}$$

$$\begin{aligned} = \sqrt{2} \epsilon \psi + 2i \theta \sigma^{\mu} \bar{\epsilon} \partial_{\mu} \psi - \sqrt{2} i (\theta \theta) (\partial_{\mu} \psi \sigma^{\mu} \bar{\epsilon}) \\ - 2 \epsilon \theta F \end{aligned}$$

$$= \delta \psi + \sqrt{2} \epsilon \psi - \theta \theta \epsilon F$$

$$\delta\phi = \sqrt{2} \epsilon \psi$$

$$\delta\psi_\alpha = \sqrt{2} i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\epsilon}^{\dot{\alpha}} \partial_\mu \phi - \sqrt{2} \epsilon_\alpha F$$

$$\delta F = \sqrt{2} i (\partial_\mu \psi \sigma^\mu \bar{\epsilon})$$

Let $\sqrt{2} \psi := \frac{1}{\sqrt{2}} \psi'$, then we obtain
for $\chi = \begin{pmatrix} \psi_\alpha \\ \bar{\chi}^{\dot{\alpha}} \end{pmatrix}$

$$(P_L \chi)_\alpha = \psi_\alpha \quad P_R \chi = \bar{\chi}^{\dot{\alpha}}$$

Supergravity Normalization:

$$\sqrt{2} \psi =: \frac{1}{\sqrt{2}} (P_L \chi), \quad F \rightarrow -F$$

$$\delta\phi = \frac{1}{\sqrt{2}} \epsilon P_L \chi$$

$$\delta P_L \chi = \frac{1}{\sqrt{2}} (i \sigma^\mu \bar{\epsilon})_\alpha \partial_\mu \phi + \frac{1}{\sqrt{2}} \epsilon_\alpha F$$

$$\delta F = i \frac{1}{\sqrt{2}} \partial_\mu P_L \chi (\sigma^\mu \bar{\epsilon})$$

(Supergravity, $N=1$)

Antichiral superfields: $D_\alpha \Phi^\dagger(x, \theta, \bar{\theta}) = 0$
 $z^\mu := x^\mu - i(\theta \sigma^\mu \bar{\theta})$

Vector / (Real) Superfields & Generalized
Gauge transformations

* Require V^m in $\Phi(x, \theta, \bar{\theta})$ to be real.

$$\therefore \boxed{V^\dagger = V}$$

$$V(x, \theta, \bar{\theta}) = C(x) + \theta \phi + \bar{\theta} \bar{\chi} + \theta \theta M(x) + \bar{\theta} \bar{\theta} N(x) \\ + \theta \sigma^\mu \bar{\theta} V_\mu + \theta \theta \bar{\theta} \lambda + \bar{\theta} \bar{\theta} (\theta \rho) + (\theta \theta) (\bar{\theta} \bar{\rho}) + \theta^2 \bar{\theta}^2 D$$

$$\Leftrightarrow V^\dagger = V$$

$$V^\dagger = C^* + \bar{\theta} \bar{\psi} + \theta \chi + \bar{\theta} \bar{\theta} M^*(x) + \theta \theta N^*(x) \\ + (\theta \sigma^\mu \bar{\theta}) V_\mu^* + \bar{\theta} \bar{\theta} (\theta \lambda) + (\theta \theta) (\bar{\theta} \bar{\rho}) + \theta^2 \bar{\theta}^2 D^*$$

$$\therefore C = C^* \Rightarrow C(x) \in \mathbb{R} \text{ (Real scalar field)}$$

$$\bar{\psi} = \bar{\chi} \quad \& \quad \psi = \chi$$

$$M(x) = N^*(x)$$

$$V^m = (V^m)^* \Rightarrow V^m(x) \in \mathbb{R}^D$$

$$\lambda = \rho$$

$$D = D^* \Rightarrow D(x) \in \mathbb{R} \text{ (" ")}$$

$$\therefore V(x, \theta, \bar{\theta}) = C(x) + \theta \psi + \bar{\theta} \bar{\psi} + \theta \theta M(x) + \bar{\theta} \bar{\theta} M^*(x) \\ + (\theta \sigma^\mu \bar{\theta}) V_\mu(x) + (\theta \theta) \bar{\theta} \lambda(x) + \bar{\theta} \bar{\theta} (\theta \lambda) \\ + (\theta \theta) (\bar{\theta} \bar{\rho}) D(x)$$

$$\left\{ \begin{array}{l} \phi, \quad, D(x): \text{Real scalar fields} \\ M(x): \text{Complex scalar fields} \\ \lambda, \psi: \text{Spinor fields} \\ V_\mu: \text{Real Vector field.} \end{array} \right.$$

We must work a bit to obtain (4.61) of the text.

$$\Rightarrow \delta_{\text{SUSY}} V(x, \theta, \bar{\theta})$$

$$\Rightarrow \theta \theta \bar{\theta} \partial \psi(x) - \bar{\theta} \partial \psi(x) = \frac{i}{2} (\partial_n \lambda \sigma^n \bar{E} - \partial_n \bar{\lambda} \sigma^n E)$$

* The product of a Left & Right handed Super field is a Vector-Superfield:

$$\Phi^\dagger \Phi :=: \bar{\Phi} \Phi \quad \text{Since}$$

$$(\Phi^\dagger \Phi)^\dagger = \Phi^\dagger (\Phi^\dagger)^\dagger = \bar{\Phi} \Phi \quad \checkmark$$

* The sum of L & R-handed Chiral Superfields is again a Vector Superfield:

$$(\Phi + \Phi^\dagger)^\dagger = \Phi + \Phi^\dagger \quad \checkmark$$

This sum can be written as

$$\begin{aligned} \Phi + \Phi^\dagger = & \phi + \phi^* + \sqrt{2} \theta \psi(x) + (\theta \theta) F + \bar{\theta} \bar{\theta} \bar{F} \\ & + i (\theta \sigma^n \bar{\theta}) \partial_n (\phi - \phi^*) \\ & - i/\sqrt{2} (\theta \theta) \bar{\theta} \bar{\sigma}^n \partial_n \psi(x) - \frac{i}{\sqrt{2}} (\bar{\theta} \bar{\theta}) \theta \sigma^n \partial_n \bar{\psi} \\ & - \frac{1}{4} (\theta \theta) (\bar{\theta} \bar{\theta}) \square (\phi + \phi^*)(x) \end{aligned}$$

Where $V_m = \partial_m (\phi - \phi^*)$

Naturally, we'd like this to be (Super)gauge invariant. I.e. $V_m \mapsto V_m + \partial_m \Lambda$

So, we make

$$\begin{aligned} \lambda(x) &\mapsto \lambda(x) + \frac{i}{2} \theta \sigma^n \partial_n \lambda(x) \\ D(x) &\mapsto D(x) - \frac{1}{4} \square C \end{aligned}$$

This, alongside $M \mapsto \frac{i}{2} (M + iN)$ yields the expansion (4.61)

Vector multiplet:

$\phi(x)$: real scalar

$\psi(x)$: Complex - two spinor

M : Complex scalar

V_μ : Real vector

λ : complex spinor

D : Real scalar

$$\mathbb{R}_B \oplus \mathbb{R}_F$$

$$\begin{array}{c} 1_B \\ 4_F \\ 2_B \\ 4_0 \\ 4_F \\ 1_B \end{array}$$

Supergravity:

$$\phi(x) \mapsto \mathbb{C}(x)$$

$$\psi(x) \mapsto \zeta(x)$$

$$M(x) \mapsto \mathcal{H}(x)$$

$$V_\mu(x) \mapsto B_\mu(x)$$

$$\lambda(x) \mapsto P_R \lambda$$

$$D(x) \mapsto D(x)$$

* After Gauge fixing we will reduce the # of off-shell D.O.F. to $4_0 + 4_F$

$$\text{on-shell: } 2_B + 2_F.$$

$$\text{From above } V \mapsto V + (\Phi + \Phi^\dagger)$$

$$\Leftrightarrow V^\mu \mapsto V^\mu - \partial^\mu (2 \text{Im}(\Phi))$$

$$(ia - b - ia - b = -2b = -2\text{Im}(\Phi))$$

which is an abelian gauge transformation.

$$\therefore F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$$

is super-gauge invariant.

$\Rightarrow$ We define: Supersymmetric gauge transformation

$$V(x, \theta, \bar{\theta}) \mapsto V'(x, \theta, \bar{\theta}) = V(x, \theta, \bar{\theta}) + (\Phi + \Phi^\dagger) \\ \equiv V(x, \theta, \bar{\theta}) + i[\Lambda(x, \theta, \bar{\theta}) - \Lambda^\dagger(x, \theta, \bar{\theta})]$$

where $\Phi = i\Lambda$

Linear Multiplet: $\mathcal{Z} = \mathcal{Z}^\dagger$

and $D^2 \mathcal{Z} = \bar{D}^2 \mathcal{Z} = 0$

where $D^\alpha D_\alpha \equiv D^2$; $\bar{D}_{\dot{\alpha}} \bar{D}^{\dot{\alpha}} \equiv \bar{D}^2$

* $\phi: C$, $\psi = \chi \equiv \begin{Bmatrix} \end{Bmatrix}$, $M = \mathcal{H}$, $V_\mu \in \mathcal{B}_\mu$
 λ , D :

$$\mathcal{L}_D = \frac{i}{2} \partial_\mu (\epsilon \sigma^\mu \bar{\lambda} - \lambda \sigma^\mu \bar{\epsilon}) \\ = -i \partial_\mu (\lambda \sigma^\mu \bar{\epsilon} + \epsilon \sigma^\mu \bar{\lambda}) \\ = -i \partial_\mu (\lambda (\bar{\sigma}^\mu + \sigma^\mu) \bar{\epsilon})$$

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \gamma_\mu = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} \\ \gamma_\mu \gamma^\mu = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & \sigma \\ \bar{\sigma} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma \\ -\bar{\sigma} & 0 \end{pmatrix} \alpha_\alpha \begin{pmatrix} \lambda \\ \bar{\lambda} \end{pmatrix}$$

$$(\bar{\epsilon}, \bar{\lambda}) \begin{pmatrix} \sigma \lambda \\ -\sigma \bar{\lambda} \end{pmatrix} = \bar{\epsilon} \sigma \lambda - \epsilon \bar{\sigma} \bar{\lambda}$$

as above.

$$\chi \mapsto i\gamma_\mu \chi$$

$$\bar{\lambda} = \mathbb{R} \lambda,$$

$$\delta \mathcal{H} = \bar{\epsilon} \mathbb{R} \lambda =$$

$N=1$ Matter Actions:

- Note $\bar{\Phi} \Phi$ is a Vector mult. since $(\bar{\Phi} \Phi)^{\dagger} = \bar{\Phi} \Phi$

$$\begin{aligned} \Phi(y, \theta) &= \phi(y) + \sqrt{2} \theta \psi + \theta \theta F(y) \\ \Phi^{\dagger}(y, \bar{\theta}) &= \phi^*(y) + \sqrt{2} \bar{\theta} \bar{\psi} + \bar{\theta} \bar{\theta} \bar{F}(y) \\ &= \phi^*(x) - i (\theta \sigma^{\mu} \bar{\theta}) \partial_{\mu} \phi^* - \frac{1}{4} (\theta \theta) (\bar{\theta} \bar{\theta}) \square \phi^* \\ &\quad + \sqrt{2} \bar{\theta} \bar{\psi} + \bar{\theta} \bar{\theta} \bar{F}^*(x) - \frac{i}{\sqrt{2}} \theta \theta (\bar{\theta} \sigma^{\mu} \partial_{\mu} \bar{\psi}) \end{aligned}$$

$$\begin{aligned} \bar{\Phi}(y, \bar{\theta}) &= \phi(y) + i (\theta \sigma^{\mu} \bar{\theta}) \partial_{\mu} \phi - \frac{1}{4} (\theta \theta) (\bar{\theta} \bar{\theta}) \square \phi \\ &\quad + \sqrt{2} \theta \psi + \frac{i}{\sqrt{2}} (\theta \theta) \bar{\theta}_{\dot{\alpha}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha \dot{\beta}} \bar{\epsilon}^{\dot{\beta} \dot{\alpha}} + \theta \theta F \\ \therefore \bar{\Phi}^{\dagger} \Phi|_D &= \mathcal{O}(\theta^2 \bar{\theta}^2) \end{aligned}$$

$$\begin{aligned} \bar{\Phi}^{\dagger} \Phi|_D &= \theta \theta \bar{\theta} \bar{\theta} F F^* - \frac{1}{4} \theta \theta \bar{\theta}^2 \square \phi^* \phi - \frac{1}{4} \bar{\theta}^2 \theta^2 \square \phi \phi^* \\ &\quad + \left(\frac{i}{\sqrt{2}} \theta^2 \bar{\theta}_{\dot{\alpha}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha \dot{\beta}} \bar{\epsilon}^{\dot{\beta} \dot{\alpha}} \right) (\sqrt{2} \bar{\theta} \bar{\psi}) \\ &\quad - \left(\frac{i}{\sqrt{2}} \bar{\theta}^2 (\theta \sigma^{\mu} \partial_{\mu} \bar{\psi}) \right) (\sqrt{2} \theta \psi) \\ &\quad + (i \sigma^{\mu} \bar{\theta}) (\partial_{\mu} \phi^*) \theta \sigma^{\nu} \bar{\theta} \partial_{\nu} \phi \end{aligned}$$

$$\begin{aligned} &= \theta^2 \bar{\theta}^2 \left(F F^* - \frac{1}{4} \square \phi^* \phi - \frac{1}{4} \square \phi \phi^* \right) \\ &\quad + i \theta^2 \bar{\theta}_{\dot{\alpha}} \bar{\epsilon}^{\dot{\beta} \dot{\alpha}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha \dot{\beta}} \bar{\theta}_{\dot{\gamma}} \bar{\psi}^{\dot{\gamma}} \end{aligned}$$

$$\begin{aligned} \textcircled{b} &= i \theta^2 \bar{\theta}_{\dot{\alpha}} \bar{\epsilon}^{\dot{\beta} \dot{\alpha}} \bar{\theta}_{\dot{\gamma}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha \dot{\beta}} \bar{\psi}^{\dot{\gamma}} \\ \bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\gamma}} &= (a) \frac{1}{2} \bar{\epsilon}_{\dot{\alpha} \dot{\gamma}} (\bar{\theta} \bar{\theta}) \end{aligned}$$

$$= \frac{a}{2} (\delta_{\dot{\alpha}}^{\dot{\beta}} \delta_{\dot{\gamma}}^{\dot{\delta}} - \delta_{\dot{\alpha}}^{\dot{\delta}} \delta_{\dot{\gamma}}^{\dot{\beta}}) \bar{\theta}_{\dot{\beta}} \bar{\theta}_{\dot{\delta}}$$

$$= \frac{a}{2} \bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\gamma}} - \bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\gamma}}$$

$$\hookrightarrow = -a \bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\gamma}} \Rightarrow a = -1$$

$$\bar{\theta}_{\dot{\alpha}} \bar{\theta}_{\dot{\gamma}} = -\frac{1}{2} \epsilon_{\dot{\alpha}\dot{\gamma}} (\bar{\theta}^2)$$

$$\textcircled{D} = i \theta^2 \bar{\theta}^2 \left(-\frac{1}{2} \epsilon_{\dot{\alpha}\dot{\gamma}} \epsilon^{\dot{\alpha}\dot{\beta}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\gamma}} \right)$$

$$= \frac{i}{2} \theta^2 \bar{\theta}^2 \delta_{\dot{\gamma}}^{\dot{\beta}} \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\gamma}}$$

$$= \frac{i}{2} \theta^2 \bar{\theta}^2 \partial_{\mu} \psi^{\alpha} \sigma^{\mu}_{\alpha\dot{\gamma}} \bar{\psi}^{\dot{\gamma}}$$

like was for $\textcircled{C}$

$$\textcircled{D} = (\theta \sigma^{\mu} \bar{\theta}) (\theta \sigma^{\nu} \bar{\theta}) (\partial_{\mu} \phi^{\kappa}) \partial_{\nu} \phi$$

$$\int d^4x \int d^4\theta \mathbb{E}^{\dagger} \mathbb{E} = \int d^4x \left[\partial_{\mu} \phi^{\kappa} \partial^{\mu} \phi + \frac{i}{2} (\psi \sigma^{\mu} \bar{\psi}) \bar{\psi} \right]$$

$$+ F F^{\dagger} + \partial_{\mu} (1)$$

$$\int d\theta_{\alpha} = D_{\alpha}$$

$\Rightarrow$ Note: Since $\mathbb{E}$ is chiral, $\bar{D}_{\dot{\alpha}} \mathbb{E} = 0$

$$\int d^2\theta d^2\bar{\theta} \mathbb{E}^{\dagger} \mathbb{E} = \frac{1}{8} D^2 \bar{D}^2 \mathbb{E}^{\dagger} \mathbb{E}$$

$$\text{Since } [D, \bar{D}] = 0$$

$$\frac{1}{4} \int d^2\bar{\theta} \bar{D}^2 (\mathbb{E}^{\dagger} \mathbb{E}) = \frac{1}{4} \int d^2\bar{\theta} \mathbb{E}^{\dagger} D^2 \mathbb{E}$$

$$\therefore S = \frac{1}{4} \int d^4x \int d^2\bar{\theta} \mathbb{E}^{\dagger} D^2 \mathbb{E}$$

$$\delta S / \delta \mathbb{E} = 0 \Leftrightarrow D^2 \mathbb{E} = 0$$

$$\boxed{D^2 \mathbb{E} = 0}$$

$D^2 \mathbb{E} = 0 \Leftrightarrow$ Equations of Motion for

(ϕ, ψ, F)

$$\epsilon^{\alpha\beta} D_{\beta} D_{\alpha} \mathbb{E}$$

In y, θ coordinates:

$$\epsilon^{\alpha\beta} \partial_{\beta} \partial_{\alpha} \mathbb{E}$$

$$D_\alpha \Phi = \left(2i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} \phi + \sqrt{2} \psi_\alpha - 2\theta_\alpha F \right. \\ \left. + 2i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^\beta \frac{\partial}{\partial y^\mu} \psi_\beta \right. \\ \left. + \theta^\beta 2i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} (F) \right)$$

$$D_\beta (\cdot) = \left(\partial_\beta + 2i \sigma^\mu_{\beta\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} \right) (\cdot)$$

$$= -2 \epsilon_{\alpha\beta} F - 2i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \delta_\beta^\gamma \frac{\partial}{\partial y^\mu} \psi_\gamma \\ = 4i \theta_\beta \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^\mu} F \dots$$

Take a superpotential $W[\Phi]$

Take $W[\Phi]$ and expand at $\Phi = \phi$

$$W[\Phi] = W(\phi) + \left(\right) \frac{\partial W}{\partial \Phi} \Big|_{\Phi=\phi} + \left(\right)^2 \frac{\partial^2 W}{\partial \Phi^2} \\ = W(\phi) + (\sqrt{2} \theta F + \theta \theta F) \frac{\partial W}{\partial \phi} + \theta \phi \theta \phi \frac{\partial^2 W}{\partial \phi^2}$$

$$\therefore W[\Phi] = W(\phi) + \sqrt{2} \theta F \frac{\partial W}{\partial \phi} + \theta \theta \left(\frac{\partial W}{\partial \phi} F - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^2} \psi \psi \right)$$

$$\therefore \int d^2 \theta W[\Phi] = \frac{\partial W}{\partial \phi} F - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^2} \psi \psi$$

$$\text{In general, } W[\Phi] = \sum_{n=1}^3 a_n \Phi^n$$

Non-linear Sigma-model II:

Promote the complexified Gauge coupling τ to a holomorphic f^h of chiral superfields

$$\tau \int d^2 \theta \text{Tr}(W^\alpha W_\alpha) \mapsto \int d^2 \theta f_{AB}(\Phi) W^{\alpha A} W_\alpha^B$$

$$f_{AB} : \text{transforms in } \text{Adj} \otimes \text{Adj} \sim \overline{\mathbb{F}} \otimes \overline{\mathbb{F}}$$

• For Global SUSY, $f_{AB} := \tau \text{Tr}(T_A T_B)$

* Matter Fields: Those which are charged, the gauge inv. action becomes

$$K(\Phi, \bar{\Phi}) \mapsto K(\Phi, (\bar{\Phi} e^{2gV})_i) \quad \left(\begin{matrix} \text{V, in WZ} \\ \text{Gauge} \end{matrix} \right)$$

The General Action in Superspace is therefore

$$\mathcal{L} = \frac{1}{32\pi} \text{Im} \left[\int d^2\theta f_{AB} W^{\alpha A} W_{\alpha}^B \right] + \int d^2\theta d^2\bar{\theta} \left[K(\Phi^i, (\bar{\Phi} e^{2gV})_i) \right] + \int d^2\theta W(\Phi) + \int d^2\bar{\theta} \bar{W}(\bar{\Phi})$$

* The Yukawa coupling terms are now valued on the Kähler manifold

$$\bar{\Phi} \lambda \psi \mapsto K^i_j \bar{\Phi} \lambda \psi^j$$

$$K^i_j \bar{\Phi} \lambda \psi^j = K^i_j \bar{\Phi} T^a_R \lambda_a \psi^j = K^i_j (\bar{\Phi}_i |_M (T^a_R)^M_N \lambda_a (\psi^j)^N$$

$$\int d^2\theta d^2\bar{\theta} K(\Phi, \bar{\Phi} e^{2gV})$$

N=1 Super Yang Mills: G = U(1) QED.

The Superfields

$$W_{\alpha} := -\frac{1}{4} \bar{D}^2 D_{\alpha} V$$

$$\bar{W}_{\dot{\alpha}} := -\frac{1}{4} D^2 \bar{D}_{\dot{\alpha}} V$$

Note W_{α} is a superfield since V is a superfield

$$\text{and } [D_{\alpha}, Q_{\beta}] = 0$$

Also $\bar{D}_{\dot{\alpha}} W_{\alpha} = 0$ and $D_{\alpha} \bar{W}_{\dot{\alpha}} = 0$

$$\bar{D}_{\dot{\beta}} W_{\alpha} = \bar{D}_{\dot{\beta}} \left(-\frac{1}{4} \bar{D}^2 D_{\alpha} V \right) = 0$$

Since $\bar{D}^{(3)} = 0$

Calculate W_{α} in the WZ-gauge:

$$y^{\mu} := x^{\mu} + i \theta \sigma^{\mu} \bar{\theta} \quad | \quad z^{\mu} := x^{\mu} - i \theta \sigma^{\mu} \bar{\theta}$$

$$V_{WZ} = (\theta \sigma^{\mu} \bar{\theta}) A_{\mu}(y) + \theta \bar{\theta} \bar{\lambda}(y) + \theta^2 \bar{\theta} \lambda(y) + \theta^2 \bar{\theta}^2 D(y)$$

$$x^{\mu} = y^{\mu} - i \theta \sigma^{\mu} \bar{\theta} = z^{\mu} + i \theta \sigma^{\mu} \bar{\theta}$$

$$\begin{aligned} V_{WZ}(x, \theta, \bar{\theta}) &= \theta \sigma^{\mu} \bar{\theta} A_{\mu}(y^{\mu}) (i \theta \sigma^{\mu} \bar{\theta}) + \theta \bar{\theta} \bar{\lambda}(y) \\ &= (\theta \sigma^{\mu} \bar{\theta}) A_{\mu}(y) - i \theta \sigma^{\mu} \bar{\theta} (\partial_{\mu} A_{\nu}) (\theta \sigma^{\mu} \bar{\theta}) \\ &\quad + \theta^2 \bar{\theta} (\lambda(y)) + \bar{\theta}^2 \theta \lambda(y) \\ &\quad + \theta^2 \bar{\theta}^2 D(y) \end{aligned}$$

$$\begin{aligned} (\theta \sigma^{\mu} \bar{\theta}) (\theta \sigma^{\nu} \bar{\theta}) &= -\left(\frac{1}{2} \varepsilon^{\alpha\beta} \theta^2\right) \left(\frac{1}{2} \varepsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2\right) \sigma^{\mu}_{\alpha\dot{\beta}} \sigma^{\nu}_{\beta\dot{\alpha}} \partial_{\mu} \\ &= -\frac{i}{4} \theta^2 \bar{\theta}^2 \left(2 \eta^{\mu\nu} \partial_{\mu} A_{\nu} \right) \end{aligned}$$

$$V_{WZ} = \theta \sigma^{\mu} \bar{\theta} A_{\mu}(y) + \theta^2 \bar{\theta} \bar{\lambda}(y) + \bar{\theta}^2 \theta \lambda(y) + \bar{\theta}^2 \theta^2 \left(D(y) - \frac{i}{2} \partial_{\mu} V^{\mu}(y) \right) = V_{WZ}^{(1)}(x, \theta, \bar{\theta})$$

$$\left[\begin{aligned} \Rightarrow D_{\alpha}^{(1)} &= \partial_{\alpha} + 2i \sigma_{\alpha\dot{\alpha}}^{\mu} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^{\mu}} \\ D_{\dot{\alpha}}^{(1)} &= -\bar{\partial}_{\dot{\alpha}} \end{aligned} \right]$$

$$\begin{aligned} W_{\alpha} &= -\frac{1}{4} \bar{D}_{\dot{\alpha}}^{(1)} \bar{D}^{(1)} D_{\alpha}^{(1)} V_{WZ} \\ &= -\frac{1}{4} \varepsilon^{\dot{\alpha}\beta} \bar{\theta}_{\dot{\alpha}} \bar{\partial}_{\beta} \left(\partial_{\alpha} + 2i \sigma_{\alpha\dot{\alpha}}^{\mu} \bar{\theta}^{\dot{\alpha}} \frac{\partial}{\partial y^{\mu}} \right) V_{WZ} \end{aligned}$$

$$\begin{aligned} D_{\alpha}^{(1)} V^{(1)} &= \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \partial_{\mu} A_{\nu} + 2 \theta_{\alpha} \bar{\theta} \lambda + \bar{\theta}^2 \lambda_{\alpha} \\ &\quad + 2 \theta_{\alpha} \bar{\theta}^2 \left(D(y) - \frac{i}{2} \partial_{\mu} V^{\mu}(y) \right) \\ &\quad + (2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}}) (\theta \sigma^{\mu} \bar{\theta}) \partial_{\mu} A_{\nu} \quad (a) \\ &\quad + 2i \sigma^{\mu}_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \bar{\theta}^2 \bar{\theta}_{\dot{\beta}} \partial_{\mu} \lambda^{\dot{\beta}} \quad (b) \end{aligned}$$

$$\begin{aligned}
(a) \quad & 2i \sigma^\mu_{\alpha\dot{\alpha}} \sigma^\nu_{\beta\dot{\beta}} \bar{\theta}^{\dot{\alpha}} \theta^\beta \bar{\theta}^{\dot{\beta}} \partial_\mu A_\nu \\
&= -2i \sigma^\mu_{\alpha\dot{\alpha}} \sigma^\nu_{\beta\dot{\beta}} \left(\frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} (\bar{\theta}^2) \right) \theta^\beta \partial_\mu A_\nu \\
&= -i \sigma^\mu_{\alpha\dot{\alpha}} \sigma^\nu_{\beta\dot{\beta}} \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon^{\rho\gamma} \theta_\gamma \partial_\mu A_\nu \bar{\theta}^2 \\
&= +i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\sigma}^{\nu\dot{\alpha}\gamma} \theta_\gamma \partial_\mu A_\nu \bar{\theta}^2 \\
&= i (\bar{\theta}^2) (\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\gamma \partial_\mu A_\nu \theta_\gamma
\end{aligned}$$

$$\begin{aligned}
(b) \quad & 2i \sigma^\mu_{\alpha\dot{\alpha}} \bar{\theta}^{\dot{\alpha}} \theta^2 \bar{\theta}_{\dot{\gamma}} \partial_\mu \bar{\lambda}^{\dot{\gamma}}(y) \\
&= 2i \sigma^\mu_{\alpha\dot{\alpha}} \left(\frac{1}{2} \epsilon^{\dot{\alpha}\dot{\beta}} \bar{\theta}^2 \right) \theta^2 \partial_\mu \bar{\lambda}_{\dot{\gamma}}(y) \\
&= -i \theta^2 \bar{\theta}^2 \sigma^\mu_{\alpha\dot{\alpha}} \epsilon^{\dot{\alpha}\dot{\gamma}} \partial_\mu \bar{\lambda}_{\dot{\gamma}}(y) \\
&= -i \theta^2 \bar{\theta}^2 \sigma^\mu_{\alpha\dot{\alpha}} \partial_\mu \bar{\lambda}^{\dot{\alpha}}(y)
\end{aligned}$$

$$\begin{aligned}
D_\alpha^{(1)} V^{(1)} &= \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu + 2\theta_\alpha \bar{\theta} \bar{\lambda}(y) + \bar{\theta}^2 \lambda_\alpha \\
&\quad + 2\theta_\alpha \bar{\theta}^2 (D(y) - (i/2) \partial_\mu A^\mu(y)) \\
&\quad + i \bar{\theta}^2 (\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\gamma \partial_\mu A_\nu \theta_\gamma \\
&\quad - i \theta^2 \bar{\theta}^\gamma \sigma^\mu_{\alpha\dot{\gamma}} \partial_\mu \bar{\lambda}^{\dot{\gamma}} \\
&= \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu + \bar{\theta}^2 [2\delta_\alpha^\beta D(y) - i \partial_\mu A^\mu \delta_\alpha^\beta + i (\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\beta \partial_\mu A_\nu] \theta_\beta \\
&\quad + 2\theta_\alpha \bar{\theta} \bar{\lambda}(y) + \bar{\theta}^2 \lambda_\alpha(y) \\
&= \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu(y) + 2\theta_\alpha \bar{\theta} \bar{\lambda} + \bar{\theta}^2 \lambda_\alpha \\
&\quad + \bar{\theta}^2 [2\delta_\alpha^\beta D(y) - i \delta_\alpha^\beta \eta^{\mu\nu} \partial_\mu A_\nu + i (\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\beta \partial_\mu A_\nu] \theta_\beta \\
&\hookrightarrow \bar{\theta}^2 [2\delta_\alpha^\beta D(y) + i ((\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\beta - \delta_\alpha^\beta \eta^{\mu\nu}) \partial_\mu A_\nu] \theta_\beta
\end{aligned}$$

Claim: $2(\sigma^{\mu\nu})_\alpha{}^\beta = i ((\sigma^\mu \bar{\sigma}^\nu)_\alpha{}^\beta - \delta_\alpha^\beta \eta^{\mu\nu})$

$$\begin{aligned}
D_\alpha^{(1)} V^{(1)} &= \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu + 2\theta_\alpha \bar{\theta} \bar{\lambda} + \bar{\theta}^2 \lambda_\alpha \\
&\quad + \bar{\theta}^2 [2\delta_\alpha^\beta D(y) + 2(\sigma^{\mu\nu})_\alpha{}^\beta \partial_\mu A_\nu] \theta_\beta \\
&= i \theta^2 \bar{\theta}^2 \sigma^\mu_{\alpha\dot{\beta}} \partial_\mu \bar{\lambda}^{\dot{\beta}}(y)
\end{aligned}$$

$$\begin{aligned}
\Rightarrow 2(\sigma^{\mu\nu})_\alpha{}^\beta \partial_\mu A_\nu &= (\sigma^{\mu\nu})_\alpha{}^\beta \partial_\mu A_\nu + (\sigma^{\nu\mu})_\alpha{}^\beta \partial_\nu A_\mu \\
&= (\sigma^{\mu\nu})_\alpha{}^\beta F_{\mu\nu}
\end{aligned}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$D_\alpha^{(1)} V^{(1)}(\gamma, \theta, \bar{\theta}) = \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu + 2\theta_\alpha \bar{\theta}^2 + \bar{\theta}^2 \lambda_\alpha \\ + \bar{\theta}^2 [2\delta^\mu_{\alpha\dot{\beta}} D(\gamma) + (\sigma^{\mu\nu})_\alpha{}^\beta F_{\mu\nu}] \theta_\beta \\ - i\theta^2 \bar{\theta}^2 \sigma^\mu_{\alpha\dot{\beta}} \partial_\mu \bar{\lambda}^{\dot{\beta}}$$

$$\therefore W_\alpha = -\frac{1}{4} \epsilon_{\alpha\dot{\beta}} \partial^{\dot{\beta}} \partial^{\dot{\gamma}} (D_\alpha^{(1)} V) = -\frac{1}{4} \epsilon_{\alpha\dot{\beta}} \partial^{\dot{\beta}} \partial^{\dot{\gamma}}$$

Only $\bar{\theta}^2$ survive

$$\neq \frac{1}{4} \epsilon_{\alpha\dot{\beta}} \partial^{\dot{\beta}} \partial^{\dot{\gamma}} (\bar{\theta} \bar{\theta}) = \frac{1}{4} \epsilon_{\alpha\dot{\beta}} \partial^{\dot{\beta}} \partial^{\dot{\gamma}} (\bar{\theta}_{\dot{\delta}} \bar{\theta}^{\dot{\delta}}) \\ = \frac{1}{4} (-2) \left(\frac{1}{2}\right) \delta^{\alpha\dot{\beta}}_{\dot{\gamma}\dot{\delta}} \partial^{\dot{\beta}} \partial^{\dot{\gamma}} \bar{\theta}_{\dot{\delta}} \bar{\theta}^{\dot{\delta}} \\ = \left(-\frac{1}{4}\right) \partial^{\dot{\beta}} (\delta^{\dot{\beta}}_{\dot{\gamma}} \bar{\theta}^{\dot{\gamma}} - \bar{\theta}^{\dot{\gamma}} \delta^{\dot{\beta}}_{\dot{\gamma}}) = 1$$

$$W_\alpha = [2\theta_\alpha D(\gamma) + (\sigma^{\mu\nu})_\alpha{}^\beta \theta_\beta F_{\mu\nu}] \\ - i\theta^2 \sigma^\mu_{\alpha\dot{\beta}} \partial_\mu \bar{\lambda}^{\dot{\beta}} + \lambda_\alpha$$

Claim: $\bar{D}_{\dot{\alpha}} W^{\dot{\alpha}} = D^\alpha W_\alpha$

$\int d^2\theta W^\alpha W_\alpha$ is super gauge invariant

$$\epsilon^{\alpha\beta} (2\theta_\alpha D(\gamma) + (\sigma^{\mu\nu})_\alpha{}^\beta \theta_\beta F_{\mu\nu} - i\theta^2 \sigma^\mu_{\alpha\dot{\beta}} \partial_\mu \bar{\lambda}^{\dot{\beta}} + \lambda_\alpha) \\ (2\theta_\beta D(\gamma) + (\sigma^{\mu\nu})_\beta{}^\alpha \theta_\alpha F_{\mu\nu} - i\theta^2 \sigma^\mu_{\beta\dot{\alpha}} \partial_\mu \bar{\lambda}^{\dot{\alpha}} + \lambda_\beta) \\ = \int d^2\theta \epsilon^{\alpha\beta} (\theta_\beta D(\gamma) \theta_\alpha D(\gamma) + 2\theta_\beta D(\sigma^{\mu\nu})_\alpha{}^\beta \theta_\alpha F_{\mu\nu} \\ + \dots)$$

$$[\phi \sigma^{\mu\nu} \chi] = -\chi \sigma^{\mu\nu} \phi \Leftrightarrow (\theta \sigma^{\mu\nu} \theta) = 0$$

$$\text{Tr}[\sigma^{\mu\nu} \sigma^{\rho\sigma}] = \frac{1}{2} (\eta^{\mu\rho} \eta^{\nu\sigma} - \eta^{\mu\sigma} \eta^{\nu\rho}) + \frac{i}{2} \epsilon^{\mu\nu\rho\sigma}$$

$$F_{\mu\nu} F_{\rho\sigma}$$

$$D_\alpha W^\alpha = -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} - 2i\lambda \sigma^\mu \partial_\mu \lambda + D^2 + \frac{e}{4} \epsilon^{\mu\nu\rho\sigma}$$

$$\mathcal{L}_{\text{gauge}} = -F_{\mu\nu} F^{\mu\nu} - 4i\lambda \sigma^\mu \partial_\mu \lambda + 2D^2$$

Non-abelian: $V := V_A T^A$, $A = 1, \dots, \dim(\mathfrak{g})$

Define the Gaugino superfield:

$$W_\alpha = -\frac{1}{4} \bar{D}^2 (\bar{e}^\nu D_\alpha e^\nu), \quad \bar{W}_{\dot{\alpha}} = -\frac{1}{4} D^2 (\bar{e}^\nu \bar{D}_{\dot{\alpha}} e^\nu)$$

• $e^\nu \mapsto e^{i\Lambda} e^\nu e^{-i\Lambda}$, $\Xi \mapsto -i\Lambda$

In the U(1) case:

$$V \mapsto V + \Xi + \Xi^\dagger \equiv V + i(\Lambda - \Lambda^\dagger)$$

$$W_\alpha \mapsto -\frac{1}{4} \bar{D}^2 [e^{i\Lambda} \bar{e}^\nu e^{-i\Lambda} D_\alpha (e^{i\Lambda} e^\nu e^{-i\Lambda})]$$

* Note: Λ is a chiral superfield

$$\text{so } D_\alpha \bar{e}^{-i\Lambda} = 0, \quad \bar{D}_{\dot{\alpha}} e^{i\Lambda} = 0$$

$$\begin{aligned} W'_\alpha &= -\frac{1}{4} \bar{D}^2 [e^{i\Lambda} \bar{e}^\nu e^{-i\Lambda} e^{i\Lambda} \{ (D_\alpha e^\nu) e^{-i\Lambda} + e^\nu D_\alpha e^{-i\Lambda} \}] \\ &= -\frac{1}{4} \bar{D}^2 [e^{i\Lambda} \bar{e}^\nu \{ (D_\alpha e^\nu) e^{-i\Lambda} + e^\nu D_\alpha e^{-i\Lambda} \}] \\ &= e^{i\Lambda} \left(-\frac{1}{4} \bar{D}^2 \{ \bar{e}^\nu (D_\alpha e^\nu) e^{-i\Lambda} + D_\alpha \bar{e}^\nu e^{-i\Lambda} \} \right) \\ &\quad \bar{D}^2 D_\alpha e^{-i\Lambda} = 0 \\ &= -\frac{1}{4} e^{i\Lambda} \bar{D}^2 (\bar{e}^\nu D_\alpha e^\nu) e^{-i\Lambda} \\ &= e^{i\Lambda} W_\alpha e^{-i\Lambda} \end{aligned}$$

$$W_\alpha \mapsto e^{i\Lambda} W_\alpha e^{-i\Lambda} \quad \bar{W}_{\dot{\alpha}} \mapsto e^{i\Lambda} \bar{W}_{\dot{\alpha}} e^{-i\Lambda}$$

$$\begin{aligned} W_\alpha &= -\frac{1}{4} \bar{D}^2 (\bar{e}^\nu D_\alpha e^\nu) \\ &= -\frac{1}{4} \bar{D}^2 \left((1 - v + \frac{1}{2} v^2) D_\alpha (1 + v + \frac{1}{2} v^2) \right) \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{4} \bar{D}^2 \left[(1 - V + \frac{1}{2} V^2) (D_\alpha V + \frac{1}{2} D_\alpha V^2) \right] \\
&= -\frac{1}{4} \bar{D}^2 \left[(1 - V) D_\alpha V + \frac{1}{2} D_\alpha V^2 \right] \\
&= -\frac{1}{4} \bar{D}^2 D_\alpha V + \frac{1}{4} \bar{D}^2 V D_\alpha V - \frac{1}{8} \bar{D}^2 D_\alpha V^2 \\
&= -\frac{1}{4} \bar{D}^2 D_\alpha V + \frac{1}{4} \bar{D}^2 V D_\alpha V - \frac{1}{8} \bar{D}^2 D_\alpha V \cdot V - \frac{1}{8} \bar{D}^2 V \cdot D_\alpha V \\
&= -\frac{1}{4} \bar{D}^2 D_\alpha V + \frac{1}{8} \bar{D}^2 V D_\alpha V - \frac{1}{8} \bar{D}^2 D_\alpha V \cdot V \\
&= W_\alpha + \frac{1}{8} \bar{D}^2 [V, D_\alpha V]
\end{aligned}$$

→ Calc. in (1) γ -coords

$$\begin{aligned}
V(\gamma, \theta, \bar{\theta}) &= \theta \sigma^\mu \bar{\theta} A_\mu + \theta \theta \bar{\theta} \bar{\lambda} + \bar{\theta} \bar{\theta} \theta \lambda + \theta^2 \bar{\theta}^2 (D(\gamma) - \frac{i}{2} \partial^\mu A_\mu) \\
D_\alpha V &= \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} A_\mu + 2\theta_\alpha \bar{\theta} \bar{\lambda} + \bar{\theta}^2 \lambda_\alpha \\
&\quad + \bar{\theta}^2 (2\delta_\alpha^\beta D(\gamma) + (\sigma^{\mu\nu})_\alpha^\beta F_{\mu\nu}(\gamma)) \theta_\beta \\
&\quad - i \theta^2 \bar{\theta}^2 \sigma^\mu_{\alpha\dot{\beta}} \partial_\mu \bar{\lambda}^{\dot{\beta}}
\end{aligned}$$

$$\begin{aligned}
\bar{D}^2 [V, D_\alpha V] &= (\theta \sigma^\mu \bar{\theta}) \sigma^\nu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} [A_\mu, A_\nu] \\
&\quad + \theta \theta \bar{\theta} \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} [\bar{\lambda}^\gamma, A_\mu] \\
&\quad + \frac{1}{8} \bar{D}^2 \left(\theta \sigma^\mu \bar{\theta} \sigma^\nu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} [A_\mu, A_\nu] + \theta^2 \bar{\theta} \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} [\bar{\lambda}^\gamma, A_\mu] \right) \\
&\quad + \left(\frac{1}{8} \right) (\bar{\theta}^2 \bar{\theta}^2) \left(\theta^\alpha \sigma^\mu_{\alpha\dot{\beta}} \bar{\theta}^{\dot{\beta}} \bar{\theta}^{\dot{\gamma}} \sigma^\nu_{\alpha\dot{\beta}} [A_\mu, A_\nu] + (\dots) [\bar{\lambda}^\gamma, A_\mu] \right) \\
\therefore \frac{1}{8} \bar{D}^2 [V, D_\alpha V] &= \frac{1}{2} (\sigma^{\mu\nu} \theta)_\alpha [A_\mu, A_\nu] + \frac{i}{2} \theta \theta \sigma^\mu_{\alpha\dot{\beta}} [A_\mu, \bar{\lambda}^{\dot{\beta}}]
\end{aligned}$$

$$\begin{aligned}
W_\alpha(\gamma) &= -i \lambda_\alpha(\gamma) + \theta_\alpha D(\gamma) + i (\sigma^{\mu\nu} \theta)_\alpha F_{\mu\nu} + \theta \theta (\sigma^\mu D_\mu \bar{\lambda})_\alpha \\
F_{\mu\nu} &:= \partial_\mu A_\nu - \partial_\nu A_\mu - \frac{i}{2} [A_\mu, A_\nu] \\
D_\mu &:= \partial_\mu - \frac{i}{2} [A_\mu, \cdot]
\end{aligned}$$

$$\begin{aligned}
\mathcal{L}_{SYM} &= \frac{1}{32\pi} \int d^4x \int d^2\theta \text{Tr} (W^\alpha W_\alpha) \\
&= \text{Tr} \left\{ -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - i \lambda \sigma^\mu D_\mu \lambda + \frac{1}{2} D^2 \right\}
\end{aligned}$$

$$+ \frac{e^2}{4\pi^2} g^{-1} \text{tr}(F_{\mu\nu} F^{\mu\nu})$$

$$\tilde{\mathcal{L}} = \frac{\Theta_{YM}}{2\pi} + \frac{4\pi i}{g} \quad (\text{Complexified Gauge Coupling})$$

$$F_{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\sigma\rho} F_{\sigma\rho}$$

